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ORIGINAL ARTICLE

AI-Assisted Assessment of Inquiry Skills in Socioscientific Issue Contexts

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Keywords: auto scoring | inquiry-based learning | learning assessment | socioscientific issues

ABSTRACT

Background Study: Assessing learners' inquiry-based skills is challenging as social, political, and technological dimensions must be considered. The advanced development of artificial intelligence (AI) makes it possible to address these challenges and shape the next generation of science education.

Objectives: The present study evaluated the SSI inquiry skills of students in an AI-enabled scoring environment. An AI model for socioscientific issues that can assess students' inquiry skills was developed. Responses to a learning module were collected from 1250 participants, and the open-ended responses were rated by humans in accordance with a designed rubric. The collected data were then preprocessed and used to train an AI rater that can process natural language. The effects of two hyperparameters, the dropout rate and complexity of the AI neural network, were evaluated.

Results and Conclusion: The results suggested neither of the two hyperparameters was found to strongly affect the accuracy of the AI rater. In general, the human and AI raters exhibited certain levels of agreement; however, agreement varied among rubric categories. Discrepancies were identified and are discussed both quantitatively and qualitatively.

1 | Introduction

Promoting attainment of responsible research and innovation in science education is a project focused on introducing responsible research and innovation in primary and secondary education (https://www.parrise.eu) to give young learners a sense of responsibility through socioscientific issue inquiry-based learning (SSIBL). Learners can acquire scientific knowledge through inquiry (Arnold, Kremer, and Mayer 2014); this process can be described through the inquiry-based science education framework (Constantinou, Tsivitanidou, and Rybska 2018) to facilitate citizenship education (Levinson 2018). Best practices across Europe were collected to develop learning applications and in- and pre-service training courses for educators in the field of science education with a focus on sustainability in response to public awareness. The topics included sustainable development in urban ecosystems and climate change (Eilks 2015). On the other hand, course developers have increasingly advocated for long-term learning programs, which can bridge the gap between science and society; the development of such programs can be regarded as a socioscientific issue (Wiek et al. 2012). The assessment of learner solution quality in inquiry-based learning is challenging because social, political, and technological dimensions must be considered (Miller et al. 2014). Typically, scoring rubrics are developed before students perform inquiry-based tasks (Yang and Liu 2016). Moreover, training human coders in a valid and reliable fashion is expensive (Henderson et al. 2018).

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Summary

What Is Already Known About This Topic

- Socio-scientific issue inquiry-based learning (SSIBL) can foster students' scientific literacy and poses many challenges for teachers and students, such as assessing students' performance, and providing individual teaching supports in a large scale.
- Conventional automated scoring techniques have been applied to multiple-choice assessments but have demonstrated inadequate accuracy in evaluating students' learning outcomes for ill-structured questions.
- Artificial intelligence (AI) is a technology capable of emulating human cognitive functions to execute tasks necessitating intelligence, including natural language processing, image recognition, and decision-making.

What This Paper Adds

- AI was employed to facilitate SSIBL by automating the evaluation of students' responses to open-ended queries, utilising recurrent neural networks with long short-term memory (LSTM) as the underlying AI model.
- Both quantitative and qualitative data were employed to juxtapose the human rater with the SSI-AI rater, aiming to scrutinise the precision and accuracy of the SSI-AI evaluation mechanism.
- Different categories of rubrics could impact the levels of agreement between human and SSI-AI rater.
- Some potential factors, such as uncommon responses or those pertaining to spatial relationships, could impact the semantic judgement of SSI-AI raters, thereby diminishing both its precision and accuracy.

Implications for Practice and/or Policy

- Future research could amalgamate heterogeneous datasets from diverse SSI contexts to train a more robust SSI-AI model.
- By implementing the SSI-AI model within SSIBL, educators' teaching efficacy could be augmented, enabling them to offer more targeted guidance and support to individual students.

Processing numerous responses is time-consuming, which makes it infeasible in parallel (Hopkins and King 2010). To resolve the issues and improve the processes, various technologies were applied to automatic grading, which focused on applying technologies to assign grades to students' answers (Owoc, Sawicka, and Weichbroth 2021), such as answers for open-ended questions (Sychev, Anikin, and Prokudin 2020), and students' essays (Wang, Liu, and Dong 2018). For example is keyword analysis (Huang et al. 2011), in which keywords in students' arguments were collected and compared with the argumentation system database to evaluate the skill level of students' arguments. Huang et al. (2011) performed auto grading by using linear ridge regression and kernel-based support vector machine. The basic idea is to evaluate students' works by features that were extracted using the aforementioned technology.

The advanced development of artificial intelligence (AI) makes it possible to further address these challenges and shape the next generation of science education. As defined by McCarthy (2004), one of the founders of the discipline of artificial intelligence, AI refers to the science and engineering of making intelligent machines that could function in a similar way to humans. AI has led to rapid advancements in digital learning by making predictions or decisions similarly to humans and is effective for platforms offering personalised learning guidance or customised feedback (Hwang et al. 2020). For example, long short-term memory (Wang, Liu, and Dong 2018), a neural cell that is used for natural language processing. Based on long short-term memory, bidirectional encoder representations from transformers (BERT) was developed to develop automatic grading systems. BERT is a pre-trained model that is used to process human's natural language (Devlin 2018), such as the sentence-BERT network (Ndukwe et al. 2020). The research tried to develop automatic grading from rule-based, machine learning, and deep learning perspectives. Studies, such as Sychev, Anikin, and Prokudin (2020), integrated several technologies to automatically grade learner's knowledge based on open-ended questions. AI-based educational technologies could be applied in various educational scenarios and potential benefits for education were proven. One such technology is expert systems, which are learning management systems that can facilitate pedagogical plans. For example, Hwang et al. (2020) reported that a fuzzy expert system improved learning by considering the affective and cognitive status of learners and by then selecting appropriate learning materials. Smart-agent systems can efficiently send customised comments to learners. Matsuda, Weng, and Wall (2020) employed a teachable agent that provided metacognitive scaffolding to promote learning. Both the aforementioned studies showed that AI-based systems can facilitate learning and improve learning performance. Wang and Lin (2023) considered the development of AI-based educational applications from the human-centred perspective and stated that the human condition and context should be taken into account. In contrast, Siiman et al. (2023) commented that AI-assisted qualitative analyses can be used to increase transparency by analysing the agreement between human and AI interpretations. Such a comparison would clarify the criteria used for coding qualitative data. AI prompts can also be reused inexpensively, promoting personalised guidance in learning or enhancing learning performance by helping people make decisions, such as evaluating students' responses (Yang et al. 2021). Another advantage of AI in learning includes its ability to provide consecutive feedback after grading (Owoc, Sawicka, and Weichbroth 2021) and promote personalised learning (Mohd and Shahbodin 2015). From the various perspectives of human-AI integration, Spector and Ma (2019) recommended focusing on how AI can support the development of inquiry and critical thinking skills, especially for young learners. The present study evaluated the SSI inquiry skills of students in an AI-enabled scoring environment.

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2 | Literature Review

2.1 | AI And Long Short-Term Memory

AI algorithms can simulate human cognitive operations and perform humanlike actions (Wang 2019). An AI algorithm typically has a multilayer structure known as a neural network. A neural network can extract information from raw data; each layer in the neural network processes the data with a higher level of complexity, and the information from all layers is aggregated to extract knowledge (LeCun, Bengio, and Hinton 2015). AI algorithms can also make high-level decisions by integrating different types of data (e.g., textual and graphical data). AI methods are therefore increasingly applied in various learning domains, including for automatically assessing student performance (Grivokostopoulou, Perikos, and Hatzilygeroudis 2017), performing multidimensional assessments (Maestrales et al. 2021), creating peer dialogues that can assist in problem-solving (Howard et al. 2015), and assessing scientific argumentation in science education (Wang et al. 2021).

AI algorithms interpret text through natural language processing (NLP: Sundermeyer, Schlüter, and Ney 2012). Balyan, McCarthy, and McNamara (2020) employed NLP to evaluate the difficulty level of texts. NLP can be performed with recurrent neural networks (RNNs; Sherstinsky 2020), which have been developed for decades (Unadkat, Ciocoiu, and Medsker 2001) and have drawn increasing attention due to their precision and efficiency, as evidenced by a systematic literature review of 190 studies on educational data mining (Okewu et al. 2021). Among RNNs, multilayer neural networks and single-layer neural networks were found to be the most commonly used (77/190 studies, 47.2%). The RNN structure can record historical information (Sherstinsky 2020), which makes it possible to update the current state based on past state and current input data (Yu et al. 2019).

However, RNNs suffer from vanishing gradient and gradient explosion problems (Hochreiter 1998); hence, long short-term memory (LSTM) was developed to address the issues (Hochreiter and Schmidhuber 1997). A detailed introduction to the LSTM architecture can be found in Van Houdt, Mosquera, and Nápoles (2020). Equations for illustrating the relationships between the variables could be found in Yu et al. (2019).

LSTM is particularly effective when the time lag between input events and target signals is long and is computationally more efficient than RNNs. Other networks, such as bidirectional LSTM, can also perform simultaneous forward and backward inferencing (i.e., learning from opposite directions; (Schuster and Paliwal 1997)) and thus have higher prediction accuracy than single directional RNNs do (Graves, Fernández, and Schmidhuber 2005). On the other hand, LSTM outperforms RNNs in terms of error rates (Shewalkar, Nyavanandi, and Ludwig 2019). The bidirectional LSTM showed its potential in scientific learning tasks (Aljaloud et al. 2022; Wang and Lin 2023). Therefore, we selected LSTM for the development of our AI models.

2.2 | SSIBL

The SSIBL framework aims to foster democratic citizenship through responsible research and innovation in science and technology education (Levinson 2018). This educational model emphasises the necessity for science and technology to serve social desirability, aesthetic acceptability, and sustainability goals (Amos, Knippels, and Levinson 2020; Owen, Macnaghten, and Stilgoe 2012). SSIBL's pedagogical objectives are to engage students in addressing socioscientific questions and finding solutions through research and inquiry, with instruction underpinned by responsible research and innovation principles. SSIBL is intended to enhance teacher pedagogy in terms of citizenship education, SSIs, and inquiry-based science education (Knippels and van Harskamp 2018).

SSIBL consists of three principal stages: formulation of SSI-related questions (the ask stage), exploration of these questions through inquiry (the find-out stage), and making of informed decisions for an action (the act stage). This model encourages critical thinking and focuses on developing students' capacity for ethical and sustainable decision-making (Amos, Knippels, and Levinson 2020). The key student outcomes fostered by SSIBL include functional literacy, interdisciplinary knowledge, high-order thinking, and comprehensive viewpoint integration. Empirical studies, such as that of Maryam and Suwono (2023), have demonstrated the positive effects of SSIBL on students' decision-making regarding SSIs. Suwono, Saefi, and Fachrunnisa (2021) further substantiated these findings, suggesting that SSIBL enhances students' scientific literacy and critical thinking.

Despite its proven benefits in terms of teacher development and student learning, SSIBL's complexity presents major pedagogical challenges. Teachers may be hesitant to incorporate SSIs because of these challenges and a perceived lack of support in curricular frameworks. Analyses of curricula from the United Kingdom, Sweden, and the Netherlands revealed a disparity between the promotion of inquiry-based learning and the application of SSIBL's critical ask-and-act stages (Amos, Knippels, and Levinson 2020). Teachers also struggle to concurrently integrate interdisciplinary knowledge and improve students' inquiry skills. This is compounded by resistance to dialogue-based activities, which often involve multiple disciplines and perspectives (Kilinc, Demiral, and Kartal 2017).

Teachers implementing SSIBL must adeptly blend various educational strategies with a focus on students' interdisciplinary understanding, inquiry skills, and negotiation abilities. The complexity of SSIBL can reduce a teacher's ability to effectively evaluate student performance and provide tailored guidance. Cross-disciplinary collaboration among teachers has been shown to be a promising way of enriching SSIBL practices and to lead to more comprehensive and socially responsive student inquiries (Ariza et al. 2021). However, time constraints and curricular demands may limit such collaboration.

To address these pedagogical challenges, the present study designed an AI-supported system, which seeks to augment teachers' abilities by assessing student performance in SSIBL

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and delivering automated guidance and feedback. This support can give teachers more time to devote to personalised instruction, particularly for students requiring specific assistance. In sum, implementing SSIBL requires various types of knowledge so as to effectively evaluate students' performance and provide tailored guidance. Also, time efficiency is key to providing prompt assistance and avoiding interruption in SSIBL. AI-assisted evaluation could address both challenges and promote learning.

2.3 | Inquiry Skills in SSIBL

In SSIBL, students are expected to engage in the SSI context and find reasonable solutions through inquiry activity (Knippels and van Harskamp 2018). Thus, recognising students' inquiry skills in SSIBL was a critical issue in this study. Inquiry refers to a series of activities scientists work on to construct our knowledge of the physical world (Kuo et al. 2015; NRC 2000). These activities include making observations, posing questions, planning investigations, collecting, analysing, interpreting data, proposing explanations and predictions, and communicating the findings (NRC 1996). Engaging in meaningful and productive inquiry activities, students are expected to demonstrate various cognition abilities that integrate scientific knowledge and process skills. Several policy documents (Ministry of Education 2018; NRC 1996, 2000) and relevant studies (Krajcik et al. 1998; Krajcik and Czerniak 2007; Kuo et al. 2015; Wu et al. 2015) have proposed some fundamental skills to incorporate knowledge and scientific process skills into inquiry activities, such as asking questions, forming hypotheses, designing the experiment, collecting, analysing and interpreting the data. After carefully considering the nature of the SSI context, the nature of the inquiry, the amount of learning time, the scope of the learning tasks, and the students' prior knowledge and performance relevant to the SSI context, it is impossible to autoscale students' responses to all tasks during learning. Thus, this study summarised two relevant fundamental skills listed in the MOE document: "planning and researching" and "reasoning and argumentation." "Planning and researching" refers to identifying the variables relevant to questions and setting a serious procedure to collect the data. Reasoning and argumentation are defined as using appropriate data to support the claim and connecting the evidence to their claim, including using scientific principles, concepts, or experience to reason and interpret the data. The subdimensions of each inquiry skill are described in detail in the Section 3.

Moreover, multiple perspective considerations are critical in SSI teaching and learning (Georgiou and Kyza 2023; Rauch and Radmann 2020; Sadler 2009; Zhang and Hsu 2021). The more perspectives students consider (i.e., broader social issues), the more understanding of the SSI context students can acquire to implement in the inquiry activities. In other words, students' multiple perspective considerations encourage them to think of social influences relevant to their inquiry activity. Through this process, they have the opportunity to engage in responsible inquiry activities to consider social desirability, aesthetic acceptability, and sustainability, which SSIBL stressed (Amos, Knippels, and Levinson 2020; Owen, Macnaghten, and Stilgoe 2012).

2.4 | Autoscoring

Automatic scoring can reduce the time required for large-scale assessments. Conventional automatic scoring has long been used for assessing responses to multiple-choice questions; an example is an online automatic scoring system developed by Calado, Ramos, and Jonas (2019). Arifi et al. (2016) developed an effective automatic assessment method which is based on static and dynamic analysis to measure students' programming skills. However, most automatic scoring systems are inappropriate for assessing students' performance or behaviours in a complex learning context, such as SSIBL. Indeed, learning assessments should focus not only on what learners learn but also on how they learn. Automatic scoring systems are thus only moderately useful for most learning tasks. Methods of assessing learner performance in an open learning environment require further research (Käser and Schwartz 2020).

Many companies have developed automatic scoring systems that seek to overcome the limitations to existing systems. Such systems have been developed for language learning tasks, such as essay writing and speaking tests (Dilkli 2006). However, the early systems only considered the number of words. The precision of autoscoring systems may be insufficient for context-dependent texts, especially those in the Chinese language. AI might be feasible to address the issue. The review of Ramesh and Sanampudi (2022) reported that many newer automated essay scoring systems now use neural networks. Several researchers have implemented AI or NLP techniques in autoscoring or learner feedback systems (Dilkli 2006). For example, Wu and Chien (2019) constructed an RNN-based system that detects students' emotions (including anger, neutrality, joy, and sadness) while they are talking with others. Käser and Schwartz (2020) employed AI to detect students' inquiry strategies and performance and to predict their learning achievement. Using a support vector machine with recursive feature elimination, Chen, Cui, and Chu (2020) derived 27 key behaviours exhibited by students engaging in game-based learning. Sequential patterns were constructed in one study to predict students' behaviour and performance by using LSTM (Yu et al. 2019). Focused on the classification of learners' cognitive or meta-cognitive strategies, Wang and Lin (2023) employed LSTM to evaluate self-regulated learning. Bidirectional LSTM has yielded promising results for automatic grading (Xia, Liu, and Zhang 2020; Zhu and Sun 2020). Model complexity, which is crucial to understanding deep learning models (Hu et al. 2021), is one of the considerations for choosing AI models because the sample size is limited in this study. It might be insufficient for complex models, as train complex models usually need larger datasets (Alwosheel, van Cranenburgh, and Chorus 2018). The model complexity of AI models is influenced by multiple factors, including model framework, model sizes, optimization processes and data complexity (Nakkiran et al. 2021). The model complexity in this study is based on model sizes. According to Devlin (2018), the Chinese Simplified and Traditional BERT (i.e., BERT-base for Chinese) consists of 12-layers, 768-hidden state sizes, and 110-million parameters (https://github.com/google-research/bert). In contrast, the LSTM for the SSI-AI consists of three-layers, 256-hidden state size, and 2.63 million parameters. Although it is difficult to compare LSTM and BERT in

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a precise way, generally the LSTM might be less complex than BERT based on model sizes. Understanding the target assessment construct is therefore feasible. LSTM neural networks can be made transparent, as suggested by Balasubramaniam et al. (2022). With adequate reporting, researchers can understand the construction of these systems, the data they use, and, to some extent, their functioning. In other words, researchers could test factors that might affect the performance of the AI models by modifying settings (i.e., hyper-parameters) of LSTM based on the architecture illustrated in Aljaloud et al. (2022). Such features could help to gain understanding of not only what the AI model is, but also how the AI model works (i.e., explainable AIs). For example, if dropout rate has an impact on the model performance (e.g., classification accuracy), it might suggest overfitting (Dabare et al. 2018; Srivastava et al. 2014). However, gain our understanding of LSTM does not imply improving the transparency of AI models (Ferrara and Qunbar 2022), as transparency is not included in our design. Moreover, bi-directional LSTM could outperform BERT for small datasets in classification tasks (Ezen-Can 2020). These are the reasons why bi-directional LSTM was used in this study.

Ideally, teachers should be able to immediately diagnose learners' cognition and provide appropriate guidance (Käser and Schwartz 2020). Improved automatic scoring mechanisms that can measure students' performance during learning can help teachers detect students' learning progression so they can provide more guidance (Wiley et al. 2017). Studies regarding the provision of learning feedback have often only investigated particular questions or tasks with fixed solutions (de Chiusole et al. 2020; Maniktala et al. 2020; Tacoma et al. 2020). One study analysed nonverbal behaviours to explore cognitive and socioemotional dimensions during learning (Tacoma et al. 2020). In more complex learning contexts, such as SSIBL, teachers must solve more challenges in their classrooms. Using LSTM for automatic diagnosis and feedback could reduce teachers' workload and increase teaching efficacy, enabling teachers to provide more assistance and guidance (Käser and Schwartz 2020). However, large datasets would be required for training accurate AI models such as LSTM models (Dilkli 2006). In this study, an ongoing learning module was investigated, and raters with SSI teaching experience were invited to label students' work for training a model named SSI-AI. We investigated the discrepancies between human and SSI-AI labelling from both quantitative and qualitative perspectives.

We ensured the accuracy of the SSI-AI model by optimising two hyperparameters: the number of neurons in each layer (N_{neuro}) and the dropout rate; this optimisation prevented overfitting (Matsuda, Weng, and Wall 2020) and improved the model's generalisability (Ying 2019). Hyperparameter optimisation is crucial for small datasets such as ours. Three values for N_{neuro} (30, 60, and 90) and the dropout rate (0.1, 0.3, and 0.5) were tested, corresponding to the situation in a similar study (Wang and Lin 2023) to enable comparisons. Srivastava et al. (2014) also indicated that a dropout rate of 0.5 may be optimal for various NNs and tasks. It should be noted that changing hyperparameters and observing the behaviour of AI models does not reveal how the AI makes decisions. Understanding individual AI decisions is much more challenging. The approach used in the current study does not contribute to the explainability of the AI model. Some possible methods that could enhance explainability include attention visualisation, heatmaps, activation patterns, and modifying the network design (Arras et al. 2019; Dang, Nguyen, and Nguyen 2021). However, none of these methods were used in the present study, so explainability is entirely beyond the scope of this research.

In summary, the literature indicates that AI has the potential for automatically assessing students' performance and scientific argumentation in science education, and is suitable for assessing learners' inquiry ability in SSIBL. An AI model for NLP was designed using LSTM in the present study, and the hyperparameters of the model were optimised to ensure generalisability.

2.5 | The Present Study

We aimed to explore whether an AI-enabled scoring system (i.e., SSI-AI) could evaluate students' inquiry skills for SSIs. Accordingly, this study addressed the following three research questions:

RQ1. *How do N_{neuro} and the dropout rate affect the performance of the SSI-AI model?*

RQ2. *What is the performance of the SSI-AI model when evaluating SSI?*

RQ3. *What types of student answers are difficult for the SSI-AI model to correctly evaluate?*

3 | Methodology

3.1 | Participants

A large dataset of student responses was collected within an SSI learning module entitled "Coastal line change," which has been assigned to students since 2016. The learning materials were revised slightly to ensure that they met the SSIBL framework requirements for the SSI-AI trials, and 1250 student responses to the revised module were collected. The students were at least 10th graders and at most college undergraduates, and were recruited from nine schools (37 classes) in western Taiwan; all of these schools serve middle-to-upper-income communities. None of the students had previous experience of completing SSI-based learning modules. Detailed information on the learning module can be found in the Supporting Information: Appendix A.

Five raters, one male and four female, were invited to label and assess the student responses. The raters were experienced senior high school teachers who had taught Earth Science for over 10 years and who had used the "Coastal line change" learning module in their classrooms; they were therefore familiar with the structure of the module and with students' inquiry performance in the SSI context.

3.2 | Materials

Following the framework of SSIBL$^+$, an SSIBL digital lesson comprised five stages: understanding the SSI context, asking,

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enacting, acting, and reflecting (Table 1). The five experts and the research team all considered the learning tasks to be consistent with the SSIBL$^+$ framework. After discussion, 10 tasks were designed and included in the SSIBL digital lesson. Each of the five stages involved one to three tasks. As detailed in Table 1, the students were required to respond to open-ended questions for all tasks except for Tasks 3, 4, and 10.

3.3 | Architecture of the SSI-AI Model

The basic version of LSTM (i.e., 3-layers, 256-hidden state size, and 2.63 million parameters) was selected in the present studies for two reasons. First, similar models have been applied to automatic grading/classification for various educational scenarios, including mate-cognition in science education (Wang and Lin 2023), problem-solving in mathematics education (Lin 2024), and reading comprehension (Zhang et al. 2022) recently. The results showed its potential in automatic grading. Second, using similar models makes it possible to compare results and evaluate the generalizability of the model. The initial LSTM parameters of the SSI-AI model were as follows: the maximum word length was 100, and the maximum number of words was 10,000. This is sufficient for NLP for a specific topic. The dimension for a word vector in the embedded layers was 256. Details regarding these settings can be found in Keras (2015) and Gulli and Pal (2017). The activation function of the output layer was softmax, which is common for supervised classification tasks (Nwankpa et al. 2018).

The SSI-AI model processed text for feature extraction through three steps: text segmentation, tokenisation, and zero padding. In text segmentation and tokenisation, the written text was split into units (tokens) with simple meanings, such as words and topics. The tokens were transferred into a multidimensional vector that represented different aspects of meanings. Short vectors were padded with zeros at the end. Detailed information can be found in Webster and Kit (1992).

3.4 | SSI-AI Architecture

The SSI-AI model was inspired by WISE (https://wise.berkeley.edu), which was developed by Berkeley University. WISE is a digital learning platform for conveying learning tasks for SSIBL science education to foster in-depth learning and automatically score learners' responses. First, learner answers and rater evaluations (levels of responses; LVs) were collected and stored on a MySQL server. Improper or missing data were removed, and the data were preprocessed. The SSI-AI model was then trained, tested, and validated by following a feature engineering process. Subsequently, the hyperparameters were selected; most were set to the defaults for Tensorflow and Keras, but two key hyperparameters—N_{neuro} (Russell and Norvig 2020, 29) and the dropout rate (Srivastava et al. 2014)—were optimised (Sakschewski et al. 2014; Shahsavari et al. 2017). Ten-fold cross validation was then performed to validate the SSI-AI model; specifically, the data were split into 10 subsets, and each subset was then used as the testing dataset while the others were used for training (Nti, Nyarko-Boateng, and Aning 2021). The precision, recall, f1_score, and area under the receiver operating characteristic curve (ROC-AUC) were calculated, and analysis of variance (ANOVA) was performed. Figure 1 provides an overview of the process (note that incremental learning was not performed in this study).

3.5 | Procedure

Students first completed the seven open-ended Tasks (1, 2, and 5–9) of the module independently. Once improper and missing data had been removed, 1250 students' responses were evaluated by human coders. A coding rubric was developed to ensure the reliability and validity of the human labels (Table 2). For this study, we invited five senior high school teachers as raters to develop the coding rubric for the seven open-ended tasks. After carefully discussing students' performance on the seven tasks (Table 1), students' two inquiry skills and one skill relevant to SSI learning were scored, including planning and reasoning, reasoning and argumentation, and understanding of the SSI context. Together with the five raters, we then discussed the sub-dimensions of these three skills. Students' performance levels for each sub-dimension were also recognised based on their responses to the open-ended tasks. After several recursive discussions, four sub-dimensions of inquiry skills and one regarding SSI learning were identified, and each was rated at four or five levels, except for the first listed in Table 2 ("to understand the multiple variables in the SSI context"). This item was rated on a nominal scale, whereby the student responses were sorted into categories instead of levels. The four sub-dimensions of inquiry skills were: (1) predicting the relationship between variables referring to the issue (PRV); (2) proposing a series of practical procedures to gain or generate helpful information (PPP); (3) reasoning and argumentation for their claim/decision (RGG); and (4) considering alternative reasoning/viewpoints (CAV). It should be noted that the types of inquiry activities in SSIBL could be designed as various types. Experiments are not the only way to find a solution for SSI. Thus, the definition of each sub-dimension was revised to fit the requirements of the learning tasks in this study.

Given that SSI-AI is classified as a supervised learning algorithm (Russell and Norvig 2020), the total dataset was rated by human raters and was then split into training and testing datasets. Once the rubric had been defined, the five raters each labelled 10 student responses and discussed any discrepancies to ensure they shared an understanding of the coding rubric. The raters then used this rubric to label another 30 student responses independently, and the interrater reliability was evaluated. Each response was rated by at least two raters; the responses were randomly allocated. Tasks with agreement of less than 0.85 were discussed until the raters reached a consensus. This step is crucial to generate a high-quality dataset for training SSI-AI. Finally, the remaining responses were divided among the raters, and each added label individually. A total of 16,018 labels were generated.

3.6 | Data Analyses

Two hyper-parameters that might be linked to the performance of SSI-AI were tested in this study. The first one is the number of neuros in each hidden layer (N_{neuro}). Three levels of N_{neuro}, including 30, 60, and 120 were tested. The second design factor was dropout rate (Dropout rate), which was linked to the level of overfitting.

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TABLE 1 | Framework of the SSIBL digital lesson.

SSIBL$^+$ stage	Teaching goals	Learning tasks	Responses type	Performance
Understand the SSI context	Providing scaffolds and guided questions to support students' context exploring	Task 1. To infer the coastal line change based on the reading material	Open-ended writing responses	To seek the variables related to the issue
		Task 2. To clarify the problems regarding the issue via inquiring about the software	Open-ended writing responses	To understand the multiple variables in the SSI context
Ask	Introducing four coastal protection engineering solutions in the software to students and encouraging them to rethink the problems in the issue	Task 3. To explore the software to know four coastal protection engineering solutions, including their properties, advantages, and shortcomings	Exploring the software	
		Task 4. To revise the problems that students want to deal with after the exploration	Multiple choice responses	
Enact	Asking students to manipulate the software to design their protection proposal Guiding students to analyse the appropriateness of their proposal in the context	Task 5. To manipulate the software to design a suitable and reasonable protection proposal. To analyse and evaluate the effectiveness of their proposal to solve the problems they mentioned above	Open-ended writing responses	To propose a series of practical procedures to gain or generate helpful information
Act	Providing a self-evaluating activity and asking students to reflect on their protection proposal	Task 6. To predict their protection proposal's effects on coast management	Open-ended writing responses	Reasoning and argumentation for their claim/decision
		Task 7–8. To consider their protection proposal's positive and negative influence on coast management	Open-ended writing responses	• Reasoning and argumentation for their claim/decision • To understand the multiple variables in the SSI context
		Task 9. Think about how to convince others to accept their proposal	Open-ended writing responses	Considering alternative reasoning/viewpoints
Reflect	Providing a peer assessment activity and engaging students in the negotiating process in order to promote their learning about sharing, reflecting, and debating their proposal	Task 10. To take a peer assessment activity to evaluate a peer's coastal protection proposal	Oral discourse	

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框架：研究方法流程

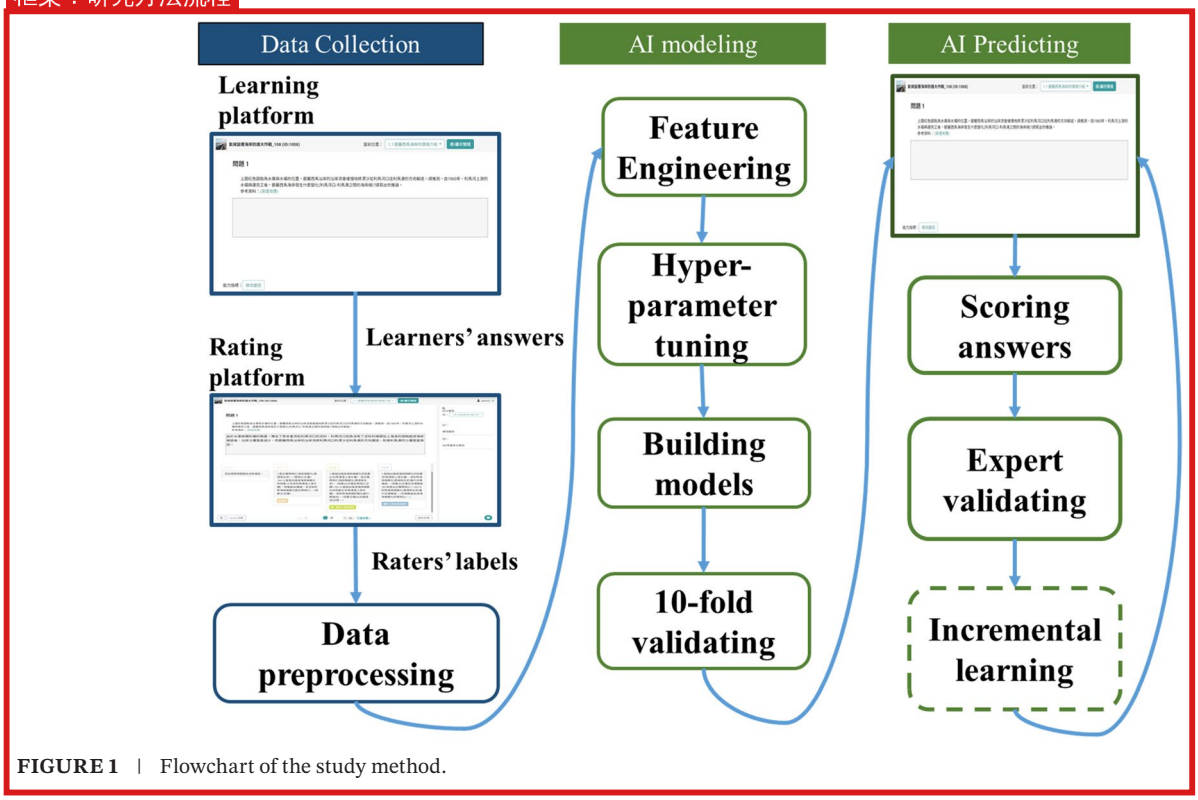


FIGURE 1 | Flowchart of the study method.

Overfitting refers to the condition in which the AI model learned too much of a training dataset, which lowers the performance of the AI model in correcting the meaning of texts retrieved from other datasets (Mutasa, Sun, and Ha 2020). By randomly dropping neuros of hidden layers, we could reduce the level of overfitting. In other words, the AI model which dropped out neuros in the hidden layers would be less sensitive to slight changes in the training samples. In the present study, three levels of dropout rate, 0.1, 0.3, and 0.5, were examined as researchers suggested that a dropout rate of up to 0.5 seems to be optimal for an AI algorithm used for general tasks (Srivastava et al. 2014). Two-way analysis of variance (ANOVA) was performed to examine whether the N_{neuro} and dropout rate had an influence on ROC-AUC.

4 | Results

4.1 | RQ1 Effects of N_{neuro} and Dropout Rate on SSI-AI Performance

The SSI-AI model was used to predict the labels for each of the five rubric items. Table 3 lists the rater scores for each item. These scores were imbalanced; the frequency of Level 0 ratings for CAV (defined in Table 2) was less than 1%. This could have affected the performance of the SSI-AI model. Therefore, item levels with a frequency of less than 1% were excluded before the AI training.

4.2 | Predicting LVs of Replies on the Basis of Each Rubric Type

The SSI-AI model was trained on each rubric item, resulting in five models. Table 4 presents the performance metrics for the five rubrics given different combinations of N_{neuro} and the dropout rate. The AUC ranged from 0.97 to 0.98.

Two-way ANOVA was performed to examine the effects of N_{neuro} and the dropout rate on the AUC. Post hoc analyses were performed using Bonferroni tests. The results presented in Table 5 suggested no significant interaction between N_{neuro} and the dropout rate [$F(4, 81) = 0.22$, $p > 0.05$] nor any significant main effects of N_{neuro} [$F(2, 81) = 0.36$, $p > 0.05$] or the dropout rate [$F(2, 81) = 2.71$, $p > 0.05$]. Hence, neither hyperparameter significantly affected the performance of SSI-UMV.

Similar analyses were performed for SSI-CAV, SSI-RGG, SSI-PRV, and SSI-PPP. The performance metrics and ANOVA results for these models are listed in Tables 4 and 5, respectively. The AUC for each model was 0.74–0.77, 0.68–0.71, 0.82–0.84, and 0.72–0.74, respectively. Our results suggested that the hyperparameters did not exhibit any significant interaction, and did not have main effects for any model except for a main effect of N_{neuro} in SSI-CAV [$F(2, 81) = 0.45$, $p < 0.001$]. That is, dropout rate did not significantly affect the performance of the SSI-AI models.

4.3 | RQ2 Performance of SSI-AI Model for Evaluating SSI

We examined the consistency of the human raters versus the SSI-AI model by calculating the mean square contingency coefficient ϕ (Todorova, Vassilev, and Surchev 2016). The ratings were significantly positively correlated ($\phi = 0.71$, $p < 0.001$), which might indicate agreement. Interrater reliability was evaluated using Cohen's k (Landis and Koch 1977). Although statistical

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TABLE 2 | Coding rubric for the SSIBL digital lesson.

Skills	Sub-dimensions	Categories/levels	Tasks
Multiple perspective consideration	Understanding the multiple variables in the SSI context (UMV) SSI	C0. Responses related to ecology C1. Responses related to residents' safety C2. Responses related to the economy C3. Responses related to traffic and transportation C4. Responses related to landscape	Task 2, Task 7, and Task 8
Planning and researching	Predicting the relationship between variables referring to the issue (PRV)	L0. Blank or incorrect responses L1. Just list the variables or predictions referring to the issue L2. Propose the variables and the predictions without a reasonable connection between them L3. Reason the relationships between variables and the predictions	Task 1
	Proposing a series of practical procedures to gain or generate helpful information (PPP)	L0. Point out unrelated problems L1. Point out related problems L2. Provide effective methods to solve problems without reasons L3. Provide effective methods to solve problems with inappropriate reasons L4. Provide effective methods to solve problems with rational reasons.	Task 5
Reasoning and argumentation	Reasoning and argumentation for their claim/decision (RGG)	L0. Blank or incorrect responses L1. Just propose the correct inference/evaluation without explanation L2. Propose the correct inference/evaluation with an inappropriate explanation L3. Propose the correct inference/evaluation with a reasonable explanation	Task 6–8
	Considering alternative reasoning/viewpoints (CAV)	L0. Blank or incorrect responses L1. Just propose alternative viewpoints without explanation L2. Reaffirm their perspective again L3. Compare the alternative viewpoint and their perspective based on the trade-off considerations	Task 9

analyses suggested that the agreement was significant, fair-to-moderate agreement was discovered (Cohen's $k = 0.27$) for the overall SSI-AI model. This suggested limited consistency between the human and SSI-AI ratings in evaluating inquiry skills in the SSI context.

Furthermore, we investigated the consistency between the human raters and SSI-AI model for each rubric item. Table 6 indicates significant consistency in terms of both Cohen's k and ϕ, respectively (ϕ, $p < 0.001$; Cohen's k, $p < 0.001$), suggesting that the automatic AI evaluation of students' arguments was related to human evaluation. However, consistency differed between rubric items. For both indicators, consistency was highest for SSI-UMV and lowest for SSI-PPP. Given that limited agreement (i.e., Cohen's $k < 0.60$) was found for most rubric items (SSI-CAV, -RGG, -PRV, and -PPP), these findings do not fully support the hypothesis in RQ2.

4.4 | RQ3 Answers Difficult for the SSI-AI Model to Correctly Evaluate

4.4.1 | Quantitative Analysis

The difficulties that the SSI-AI model had in predicting each rubric item were evaluated using a confusion matrix, which is presented in Supporting Information: Appendix B. Bold text in the table indicates answers that were correctly predicted. We found the accuracy of SSI-AI differed across the five rubric items. SSI-UMV had the highest average accuracy of over 0.85. In contrast, SSI-PPP had the

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TABLE 3 | Frequency distribution of levels for each rubric.

Rubrics	Sample size	Lv0	Lv1	Lv2	Lv3	Lv4
SSI-UMV	5800	1267 (21.84%)	547 (9.43%)	1819 (31.36%)	945 (16.29%)	1222 (21.07%)
SSI-CAV	754	1 (0.13%)	288 (38.20%)	272 (36.7%)	193 (25.60%)	Null
SSI-RGG	4043	778 (19.24%)	868 (21.47%)	1447 (35.79%)	950 (23.50%)	Null
SSI-PRV	1253	377 (30.9%)	180 (14.37%)	481 (38.39%)	215 (17.16%)	Null
SSI-PPP	4168	889 (21.57%)	729 (17.49%)	374 (8.97%)	234 (5.61%)	19.23 (46.35%)

lowest average accuracy of below 0.30. In other words, some rubrics were easier to predict for SSI-AI, while others were not.

4.4.2 | Qualitative Analysis

The inconsistencies between the human and SSI-AI ratings were evaluated by the human raters, and some examples are listed in Table 7. The experts identified deficiencies in the SSI-AI model's semantic judgement; its errors were attributed to its inability to recognise some words and phrases in the responses, especially for responses with a small sample size. Moreover, the SSI-AI model tended to rate incomplete responses unduly highly if they contained sufficient keywords, such as in Example 1 shown in Table 7 (human rating = 1; SSI-AI rating = 2). The student meant to describe the change in the coastline but unnecessarily discussed the river; the SSI-AI model may have awarded an additional point for this erroneous statement because it contained relevant keywords ("the sand decreased"). Similarly, the SSI-AI model awarded 2 points to the completely incorrect response to the same question in Example 2. This may be because the model did not understand the spatial relationships "in front of" and "behind." The SSI-AI model was also confused if a student used advanced language (vortex, Example 3) or provided an irrelevant response (Example 4); it underrated and overrated these answers, respectively. However, the human rater labelled Example 5, an incorrect answer, as Level 2; the AI autoscoring system correctly gave a 0. This may be attributable to attentional fatigue during the extensive labelling process; the AI did not suffer from this problem.

Answer format also affected the performance of the SSI-AI model. For most responses, the students wrote their answer in a single blank area, but for the table in Task 5 (Figure 2), only the response in the last column (entitled "Reasons") was used to measure performance; the content in the first column was also relevant. The SSI-AI model seemed to consider the last column alone, which led to poor consistency.

Some tasks were measured with the same rubric item, such as Tasks 6–8 for measuring RGG. Students proposed a coastline protection design and were asked to consider its effects on coast management in Task 6, and other positive and negative influences in Tasks 7 and 8, respectively. Students' types of responses could vary greatly in these tasks, but the SSI-AI model was limited in that it considered only the rubric and not the task.

5 | Discussion

The rapid development of AI-based digital techniques enables the development of systems that can consistently and efficiently evaluate the quality of learners' arguments to facilitate SSI courses, improving science education for citizenship (Ratcliffe and Grace 2003). Computer-based learning environments can foster learner abilities in socioscientific reasoning and decision-making (Eggert et al. 2017). This study explored the potential of an AI-enabled scoring system (i.e., SSI-AI) for evaluating inquiry skills in the context of SSIs. We investigated factors that may affect the performance of the SSI-AI model and the difference between human and SSI-AI-based coding. Consistent with the findings of previous studies, bidirectional LSTM was suitable for performing automatic grading (Xia, Liu, and Zhang 2020; Zhu and Sun 2020). Neither N_{neuro} nor the dropout rate affected the AUC, suggesting that our tested values avoided model overfitting (Mutasa, Sun, and Ha 2020) and that SSI-AI has high generalisability (Ying 2019). The human and AI ratings were significantly consistent, but the agreement was limited, suggesting that the LSTM-based SSI-AI model can capture lexical meaning with chronological relations and can automatically evaluate the quality of learners' responses (Zhao, Sun, and Jin 2018). However, the accuracy might differ across rubric items. To provide more comprehensive information, we compare the present study with the other study that focused on automatic grading using other technologies. It is difficult to find studies using our dataset as it is not open to the public; therefore, we selected studies with sample sizes that are roughly equal to our dataset. One advantage is the comparison would be fairer as sample sizes are highly related to model performance. Our study is compared with Surya, Gayakwad, and Nallakaruppan (2019), in which Bi-LSTM, BERT, and CNN were used for grading short answers. Both studies are identical in the batch sizes, zero padding, first-order optimizer, and dropout rate. In addition, two studies are similar in training sample size, maximal input sequence length, embedded dimension, and evaluating metrics. Given that it would be difficult to control all settings, the comparison should be considered as being general. According to the results, the Kappa of the Bi-LSTM for SSI-AI varies between 0.87 and 0.16, while the quadratic weighted Kappa of the Bi-LSTM, BERT, and CNN in

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TABLE 4 | Means (standard deviations) of the metrics for the five rubrics.

	N_{neuro}	Dropout_rate	Precision	Recall	f1_score	ROC-AUC
SSI-UMV	30	0.1	0.88(0.03)	0.87(0.04)	0.87(0.03)	0.97(0.01)
	30	0.3	0.88(0.03)	0.87(0.03)	0.87(0.03)	0.97(0.01)
	30	0.5	0.88(0.03)	0.88(0.03)	0.88(0.03)	0.98(0.01)
	60	0.1	0.87(0.03)	0.86(0.04)	0.86(0.03)	0.97(0.01)
	60	0.3	0.88(0.03)	0.87(0.04)	0.87(0.03)	0.98(0.01)
	60	0.5	0.89(0.03)	0.89(0.03)	0.89(0.03)	0.98(0.01)
	120	0.1	0.88(0.03)	0.88(0.04)	0.88(0.03)	0.98(0.01)
	120	0.3	0.89(0.03)	0.88(0.03)	0.88(0.03)	0.98(0.01)
	120	0.5	0.89(0.03)	0.88(0.03)	0.89(0.03)	0.98(0.01)
SSI-CAV	30	0.1	0.63(0.06)	0.60(0.07)	0.60(0.06)	0.76(0.07)
	30	0.3	0.61(0.09)	0.60(0.08)	0.60(0.08)	0.76(0.07)
	30	0.5	0.65(0.06)	0.63(0.06)	0.63(0.06)	0.77(0.06)
	60	0.1	0.63(0.06)	0.61(0.07)	0.61(0.07)	0.75(0.05)
	60	0.3	0.62(0.06)	0.60(0.06)	0.60(0.06)	0.75(0.06)
	60	0.5	0.62(0.07)	0.61(0.07)	0.61(0.07)	0.74(0.06)
	120	0.1	0.64(0.06)	0.61(0.07)	0.61(0.07)	0.76(0.07)
	120	0.3	0.60(0.06)	0.57(0.07)	0.57(0.07)	0.74(0.07)
	120	0.5	0.62(0.06)	0.59(0.06)	0.60(0.06)	0.75(0.07)
SSI-RGG	30	0.1	0.47(0.07)	0.46(0.07)	0.46(0.07)	0.70(0.06)
	30	0.3	0.47(0.06)	0.47(0.06)	0.46(0.06)	0.71(0.05)
	30	0.5	0.49(0.05)	0.49(0.04)	0.48(0.04)	0.73(0.05)
	60	0.1	0.48(0.07)	0.48(0.07)	0.47(0.07)	0.71(0.06)
	60	0.3	0.47(0.06)	0.47(0.05)	0.46(0.05)	0.70(0.04)
	60	0.5	0.50(0.06)	0.48(0.05)	0.48(0.05)	0.72(0.04)
	120	0.1	0.49(0.07)	0.48(0.07)	0.48(0.07)	0.71(0.05)
	120	0.3	0.48(0.07)	0.48(0.06)	0.48(0.06)	0.71(0.05)
	120	0.5	0.49(0.07)	0.49(0.06)	0.48(0.06)	0.72(0.04)
SSI-PRV	30	0.1	0.60(0.05)	0.58(0.05)	0.58(0.05)	0.83(0.02)
	30	0.3	0.59(0.06)	0.57(0.06)	0.57(0.06)	0.83(0.04)
	30	0.5	0.61(0.07)	0.58(0.06)	0.58(0.07)	0.84(0.03)
	60	0.1	0.59(0.07)	0.55(0.05)	0.55(0.06)	0.82(0.03)
	60	0.3	0.57(0.06)	0.55(0.05)	0.55(0.06)	0.82(0.03)
	60	0.5	0.60(0.06)	0.58(0.05)	0.58(0.05)	0.83(0.03)
	120	0.1	0.59(0.06)	0.57(0.06)	0.57(0.06)	0.82(0.03)
	120	0.3	0.57(0.07)	0.57(0.08)	0.56(0.07)	0.82(0.04)
	120	0.5	0.59(0.07)	0.57(0.05)	0.57(0.06)	0.83(0.03)

(Continues)

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TABLE 4 | (Continued)

	N_{neuro}	Dropout_rate	Precision	Recall	f1_score	ROC-AUC
SSI-PPP	30	0.1	0.36(0.08)	0.36(0.08)	0.35(0.08)	0.73(0.03)
	30	0.3	0.36(0.07)	0.37(0.08)	0.36(0.07)	0.74(0.02)
	30	0.5	0.36(0.08)	0.36(0.08)	0.35(0.08)	0.74(0.02)
	60	0.1	0.36(0.07)	0.37(0.08)	0.36(0.08)	0.72(0.03)
	60	0.3	0.36(0.07)	0.36(0.08)	0.35(0.08)	0.73(0.03)
	60	0.5	0.37(0.08)	0.37(0.08)	0.37(0.08)	0.74(0.03)
	120	0.1	0.36(0.08)	0.36(0.08)	0.36(0.08)	0.72(0.03)
	120	0.3	0.37(0.07)	0.36(0.07)	0.36(0.07)	0.72(0.03)
	120	0.5	0.37(0.09)	0.36(0.08)	0.36(0.08)	0.74(0.02)

結論：超參數 ANOVA（RQ1）

TABLE 5 | Results of two-way ANOVA for the five rubric items.

Items	Source	Sum of squares	*df*	Mean SS	*F*	Sig.	$\eta^2_{partial}$
SSI-UMV	N_{neuro}	4.61E-5	2	2.31E-5	0.36	0.70	0.01
	Dropout_rate	3.49E-4	2	1.75E-4	2.72	0.07	0.06
	N_{neuro} × dropout_rate	5.63E-5	4	1.41E-5	0.22	0.93	0.01
	Error	0.01	81	6.43E-5			
SSI-CAV	N_{neuro}	0.004	2	0.002	0.45	0.64	0.01
	Dropout_rate	0.001	2	0.000	0.11	0.90	0.00
	N_{neuro} × dropout_rate	0.004	4	0.001	0.26	0.91	0.01
	Error	0.321	81	0.004			
SSI-RGG	N_{neuro}	0.000	2	9.605E-5	0.04	0.96	0.00
	Dropout_rate	0.005	2	0.003	1.07	0.35	0.03
	N_{neuro} × dropout_rate	0.001	4	0.000	0.12	0.98	0.01
	Error	0.198	81	0.002			
SSI-PRV	N_{neuro}	0.001	2	0.001	0.666	0.52	0.02
	Dropout_rate	0.002	2	0.001	1.14	0.32	0.03
	N_{neuro} × dropout_rate	0.000	4	5.60E-5	0.06	0.99	0.00
	Error	0.079	81	0.001			
SSI-PPP	N_{neuro}	0.001	2	0.001	0.94	0.40	0.02
	Dropout_rate	0.004	2	0.002	2.92	0.06	0.07
	N_{neuro} × dropout_rate	0.001	4	0.000	0.29	0.88	0.01
	Error	0.053	81	0.001			

Surya, Gayakwad, and Nallakaruppan (2019) were 0.65, 0.71, and 0.62 respectively. The results showed that using Bi-LSTM in an SSI scenario could be promising, as the Kappa is higher than BERT or CNN for some specific rubrics (e.g., SSI-UMV: 0.87). On the other hand, rubrics could be a significant factor as low Kappa were also observed for other rubrics (e.g., SSI-RGG: 0.16 and SSI-PPP: 0.32). This result also reflected that training Bi-LSTM for different rubrics seems challenging. Our study contributes to the literature by demonstrating how AI can be supportive in the evaluation of SSI-related reasoning and inquiry skills, echoing the human-centred perspective of Spector and Ma (2019). AI-assisted qualitative analyses can increase transparency by enabling comparison of human and AI answer classifications (Siiman et al. 2023). The SSI-AI ratings were more consistent than were the human ratings in the present study. That is, the SSI-AI model tended to classify responses with minor differences (e.g., punctuation or conjunctions)

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into the same levels, whereas humans did not. We analysed the agreement between the human and SSI-AI coding for each rubric. Although agreement was high for SSI-UMV, it was lower for SSI-PPP; this may have been because of the greater variability in the SSI-PPP responses.

Although training on decision-making can enhance learners' abilities to address SSIs (Gresch, Hasselhorn, and Bögeholz 2013) and improve their content learning (Klosterman and Sadler 2010; Sadler, Romine, and Topçu 2016), evaluating decision-making is a challenge when embedding SSI into science education. As mentioned by Zeidler, Herman, and Sadler (2019), measurement instruments that can detect how educational goals could be achieved from learner responses in SSI courses are lacking. Although efforts have been made to assess socioscientific reasoning abilities by using multiple-choice questionnaires (Romine, Sadler, and Kinslow 2017), these items may poorly estimate SSI competencies. Other studies—such as that of Garrecht et al. (2020)—have emphasised assessment focused on structural (decision-making strategies) or contextual (decision content) conditions. Therefore, studies have employed open-ended questions to evaluate socioscientific reasoning and decision-making (Sakschewski et al. 2014). Although the results of these studies have suggested that the measurements had high reliability and validity, efficient testing on large samples remains challenging. This study provides a possible solution for automatic scoring and assessing inquiry skills on large samples.

結論：人機評分一致性（RQ2）

TABLE 6 | Consistency of human graders with the SSI-AI model for each rubric item.

Rubrics	ROC-AUC	ϕ coefficient	Cohen's k
SSI-UMV	0.98	1.72, $p < 0.001$	0.87, $p < 0.001$
SSI-CAV	0.80	0.79, $p < 0.001$	0.53, $p < 0.001$
SSI-RGG	0.83	0.42, $p < 0.001$	0.16, $p < 0.001$
SSI-PRV	0.86	0.93, $p < 0.001$	0.51, $p < 0.001$
SSI-PPP	0.79	0.81, $p < 0.001$	0.32, $p < 0.001$

5.1 | Limitations

This study has several limitations. First, the number of responses and levels for each rubric item was imbalanced. Therefore, the model favoured larger classes. Future studies could address this bias. Moreover, AI methods may not be effective for self-regulated learning for more complex scientific tasks. Although our SSI-AI model seems promising for evaluating the quality of learners' solutions for inquiry-based tasks, context analysis remains challenging because the SSI-AI model used an inductive (i.e., data-driven) approach, which was sometimes confused by surface structures (lexical levels). Moreover, such an analysis can offer only general summaries, not new insights (Graneheim, Lindgren, and Lundman 2017). In the future, researchers could develop smarter AIs that are capable of assessing higher-order strategies associated with solutions (Osborne 2013).

5.2 | Implications and Future Research Suggestions

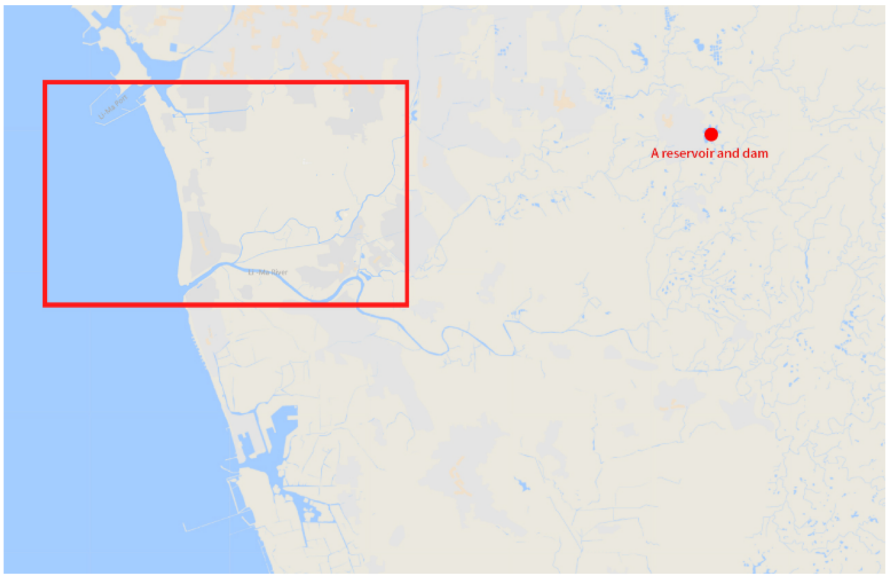
Socioscientific reasoning can be transferred across contexts (Romine et al. 2020); hence, our SSI-AI model could be

結論：人機差異原因（RQ3）

TABLE 7 | Potential causes of differences between human and SSI-AI raters.

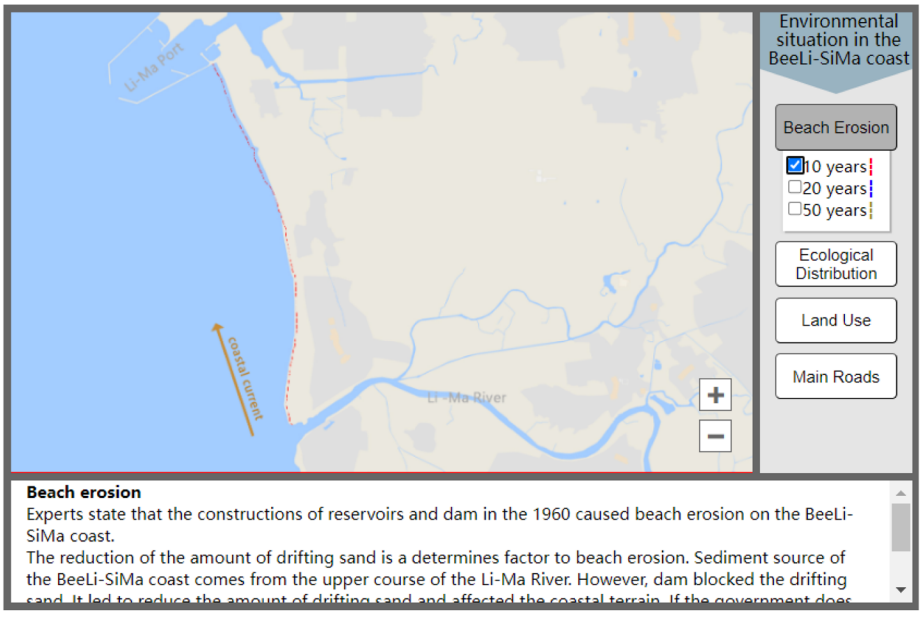
Causes	Descriptions	Rating results
1. Partially correct responses/keywords	Example 1 in Task 1. "The sand near the Li-Ma river will decrease. The sand would be accumulated near the Li-Ma harbour to influence its works"	(cid 13,045, Human: LV 1, SSI-AI: LV 2)
2. Responses consisting of spatial relationships	Example 2 in Task 1. "The coast front of the Li-Ma harbour would be eroded due to the coastal current. The coast behind the Li-Ma harbour would be regressive because of the accumulation of the sand (the coastal current was stopped by the Li-Ma harbour)"	(cid 14,298, Human: LV 0, SSI-AI: LV 2)
3. Rare/Unclear responses	(3.1) Example 3 in Task 8. "The wave dissipating block used by the revetment will destroy the coastal landscape and cause the vortex flow which will affect people's safety" (3.2) Example 4 in Task 2. "The sea turtles may be run over by cars"	(cid 11,248, Human: LV 3, SSI-AI: LV 2) (cid 3897, Human: LV 0, SSI-AI: LV 3)
4. Attentional fatigue	Example 5 in Task 1. "The reduction of river sediment accumulation leads to the retreat of the coastline"	(cid 13,099, Human: LV 2, SSI-AI: LV 0)

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Question 1
What change has the coastline from Li-Ma Port to Li-Ma estuary undergone since the reservoir construction was completed in 1960? What has the impact of these coastline changes been on the local environment?

Question 2
As shown by the environmental information in the software, what problems occur on the BeeLi-SiMa coast?



No.	Problems
1	
2	
3	
4	
5	
6	
7	

FIGURE 2 | Task 5 in the learning module.

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generalisable to other topics. Future studies could combine heterogeneous datasets for various SSI contexts and use them to train an SSI-AI model. Moreover, scholars could attempt to improve the model's accuracy by avoiding problems such as imbalanced data or rare words by combining classes with few samples or by collecting responses from specific students; for example, responses from gifted students could be used to increase the sample size for LV4 responses.

Author Contributions

Wen Xin Zhang: writing – original draft, methodology, investigation, validation, conceptualization. **John J. H. Lin:** formal analysis, methodology, conceptualization, writing – original draft, validation. **Ying-Shao Hsu:** conceptualization, writing – review and editing, funding acquisition, supervision, data curation, resources, methodology.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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Additional supporting information can be found online in the Supporting Information section.

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